US 20250268424A1

(19) United States

(12) Patent Application Publication

Davis

(10) Pub. No.: US 2025/0268424 A1

(43) Pub. Date: Aug. 28, 2025

(54) MOTORIZED GRINDER DEVICE

(71) Applicant: **Jason Davis**, Ambler, PA (US)

(72) Inventor: **Jason Davis**, Ambler, PA (US)

(21) Appl. No.: **18/588,654**

(22) Filed: **Feb. 27, 2024**

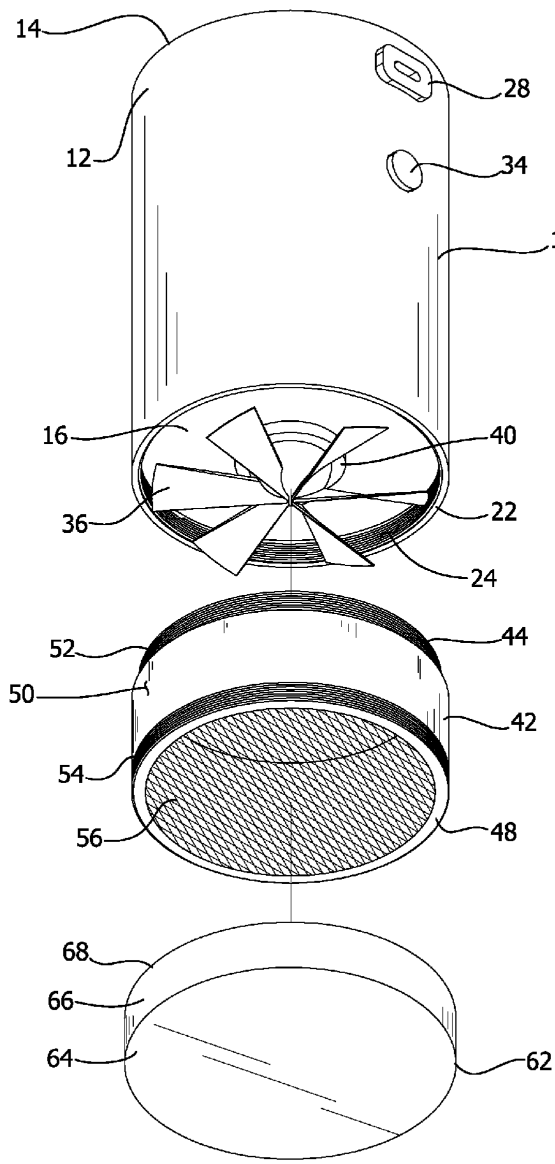
Publication Classification

(51) **Int. Cl.**
A47J 42/26 (2006.01)
A47J 42/30 (2006.01)

(52) **U.S. Cl.**
CPC *A47J 42/26* (2013.01); *A47J 42/30* (2013.01)

(57) **ABSTRACT**

A motorized grinder device for grinding herbs and other plant matter into particulates with a motorized blade and sorting the particulates by size includes a housing. A power source is positioned inside the housing. A motor is electrically coupled to the power source. The motor rotates a plurality of blades coupled to the housing for grinding an herb. A chamber is removably couplable to the housing. The plurality of blades is positioned within the chamber when the chamber is coupled to the housing so that the plurality of blades can grind the herb within the chamber. The chamber includes a mesh screen that supports the herb within the chamber. The mesh screen is foraminous to release a granulated residue of the herb from the chamber when the herb is ground. A tray is removably couplable to the chamber to catch the granulated residue of the herb.



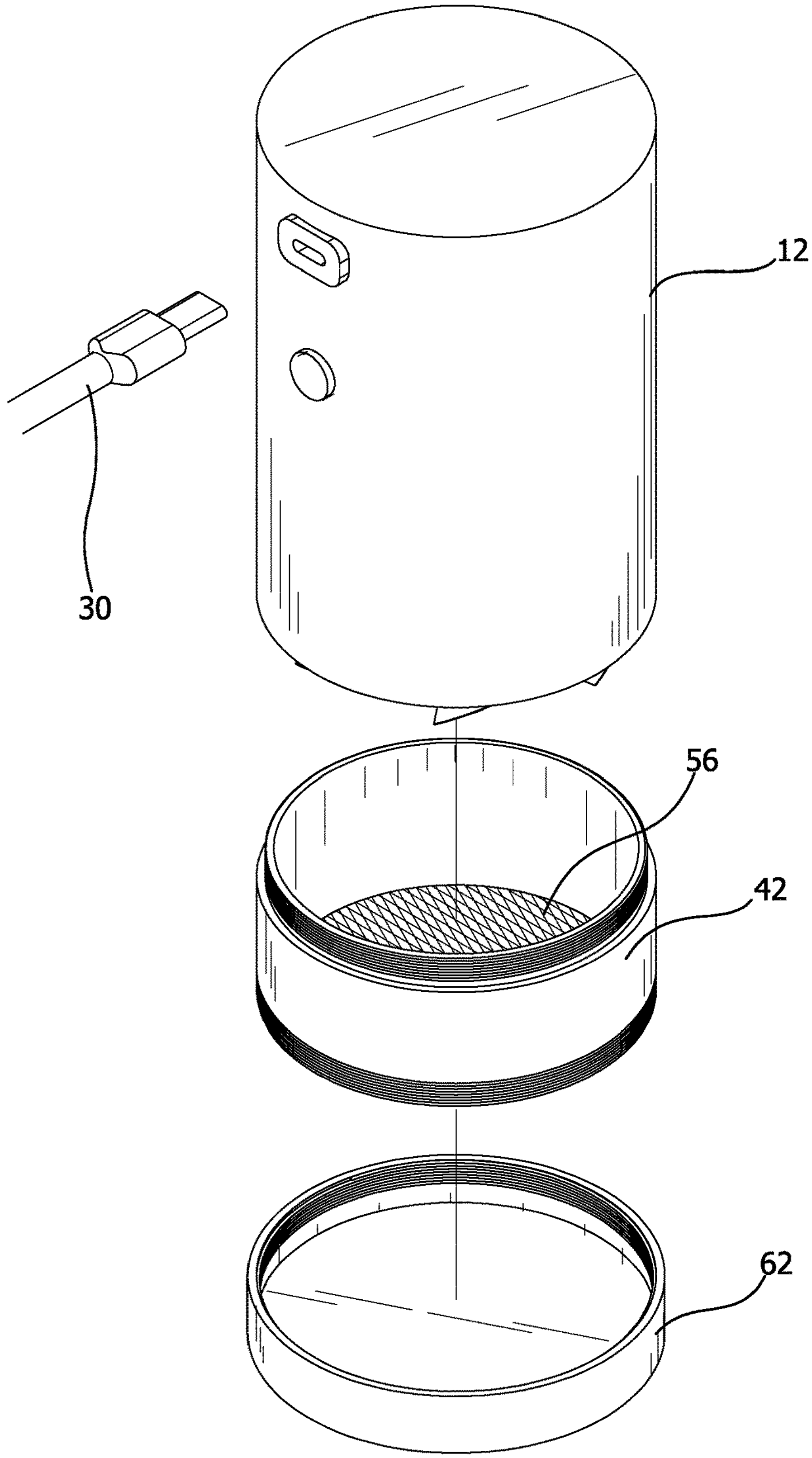


FIG. 1

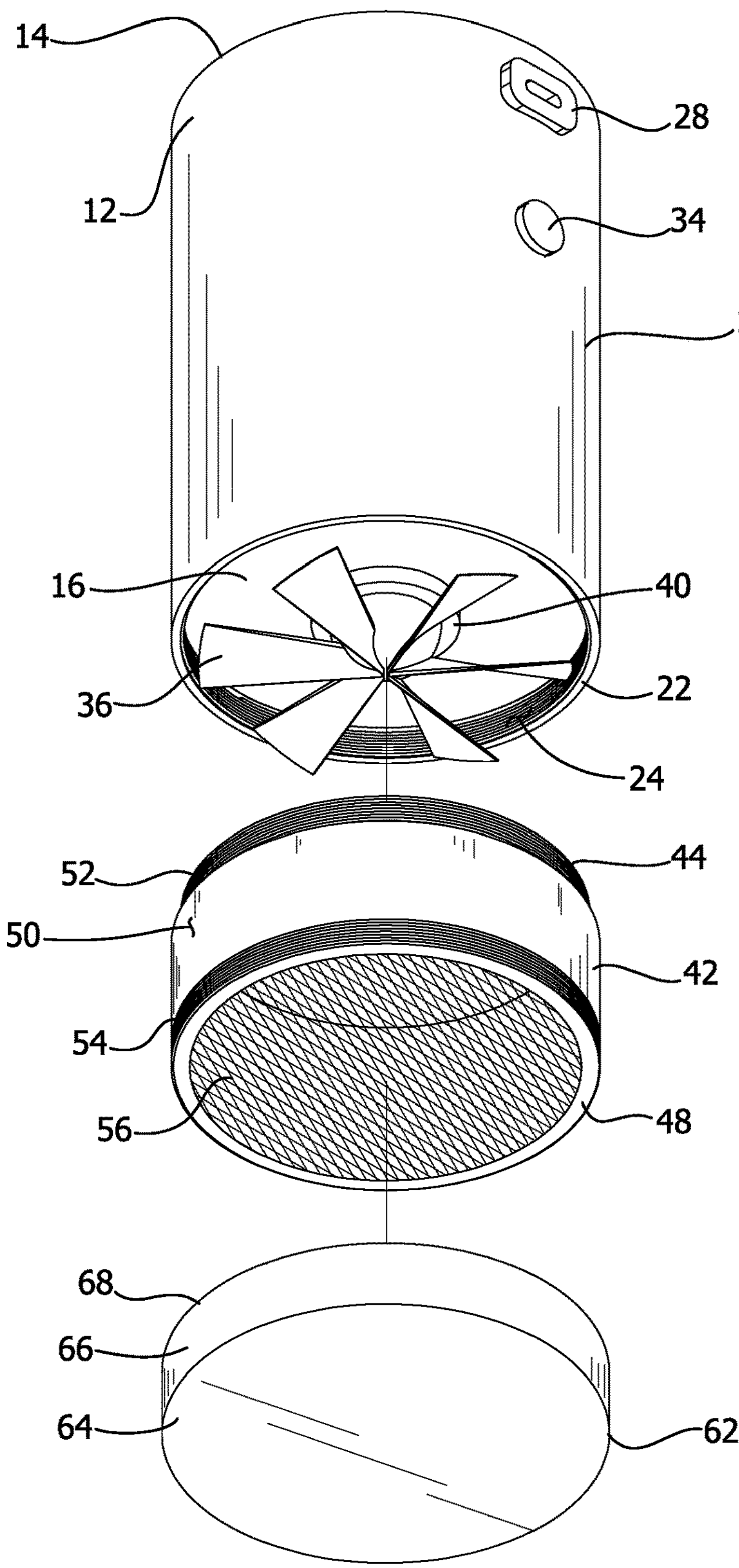


FIG. 2

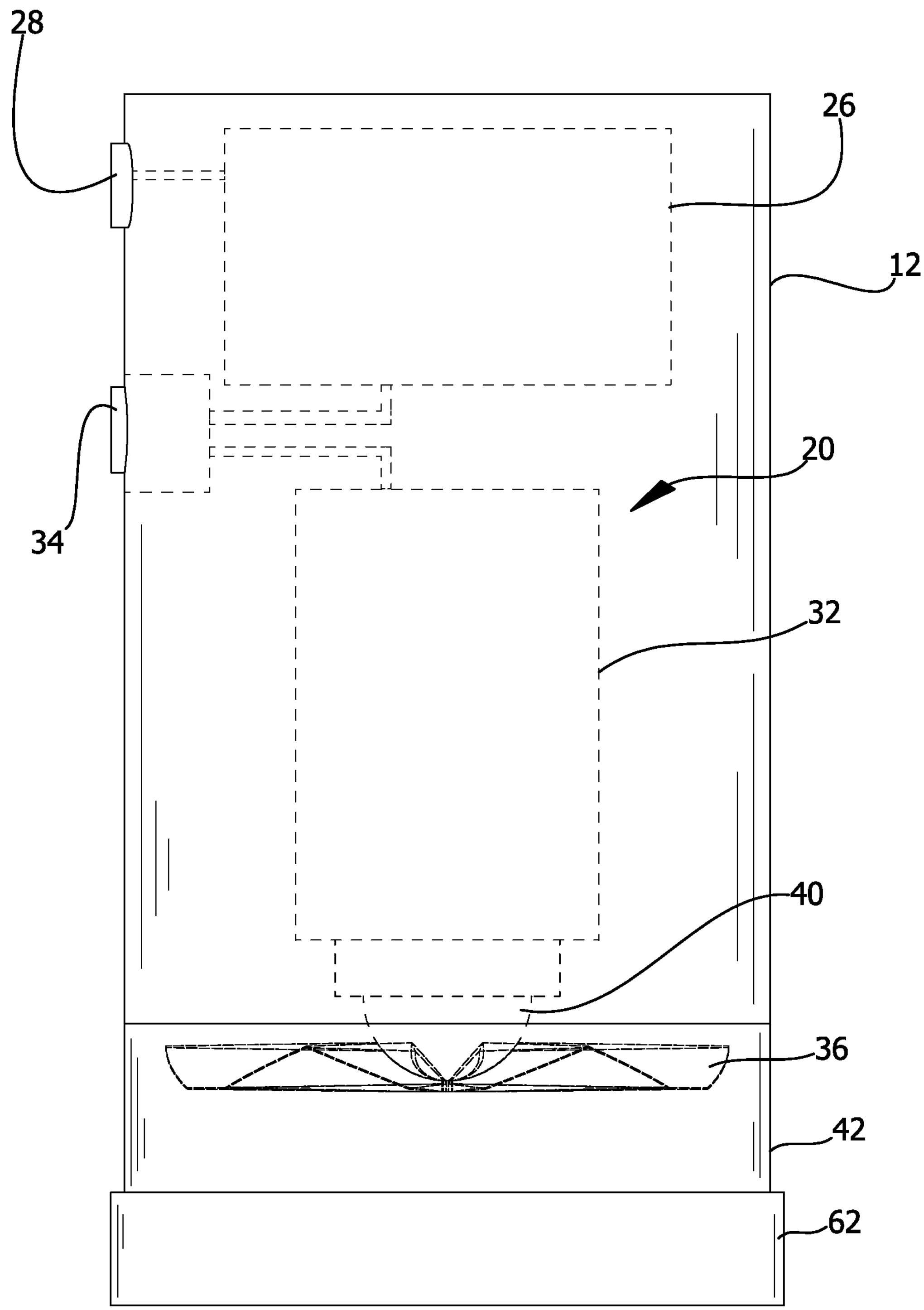


FIG. 3

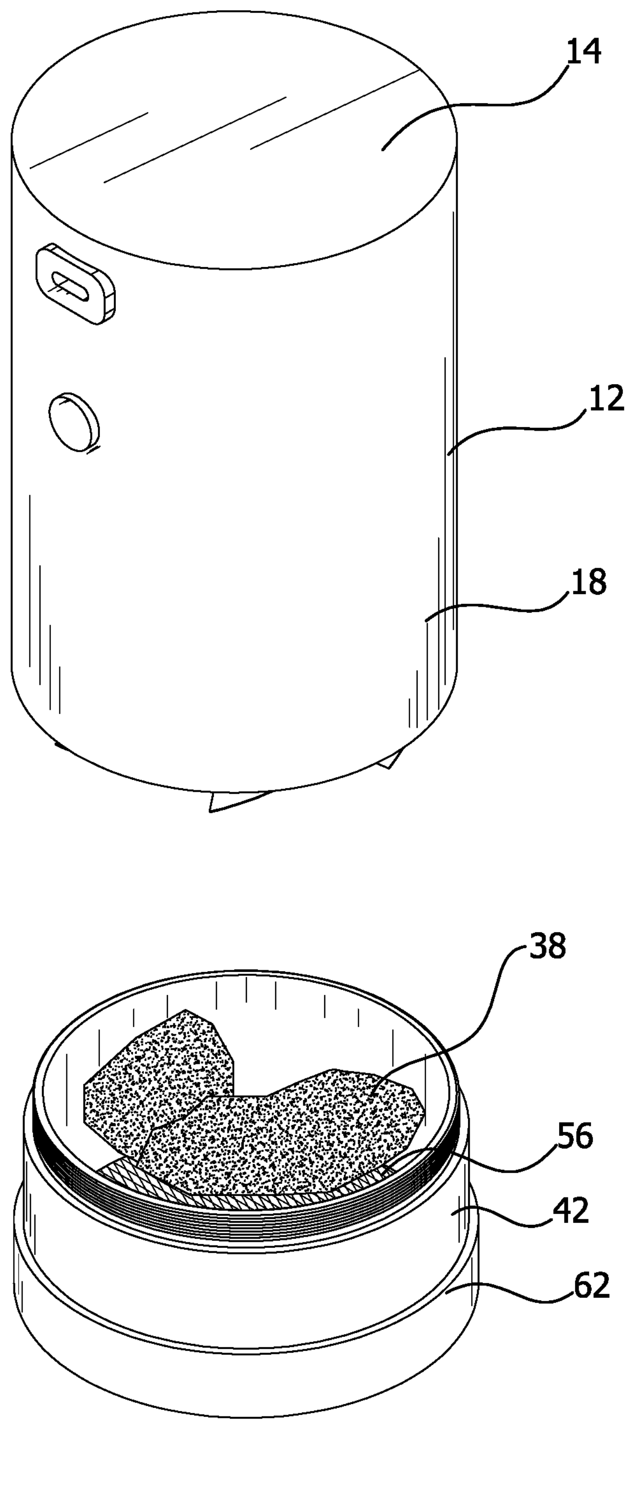


FIG. 4

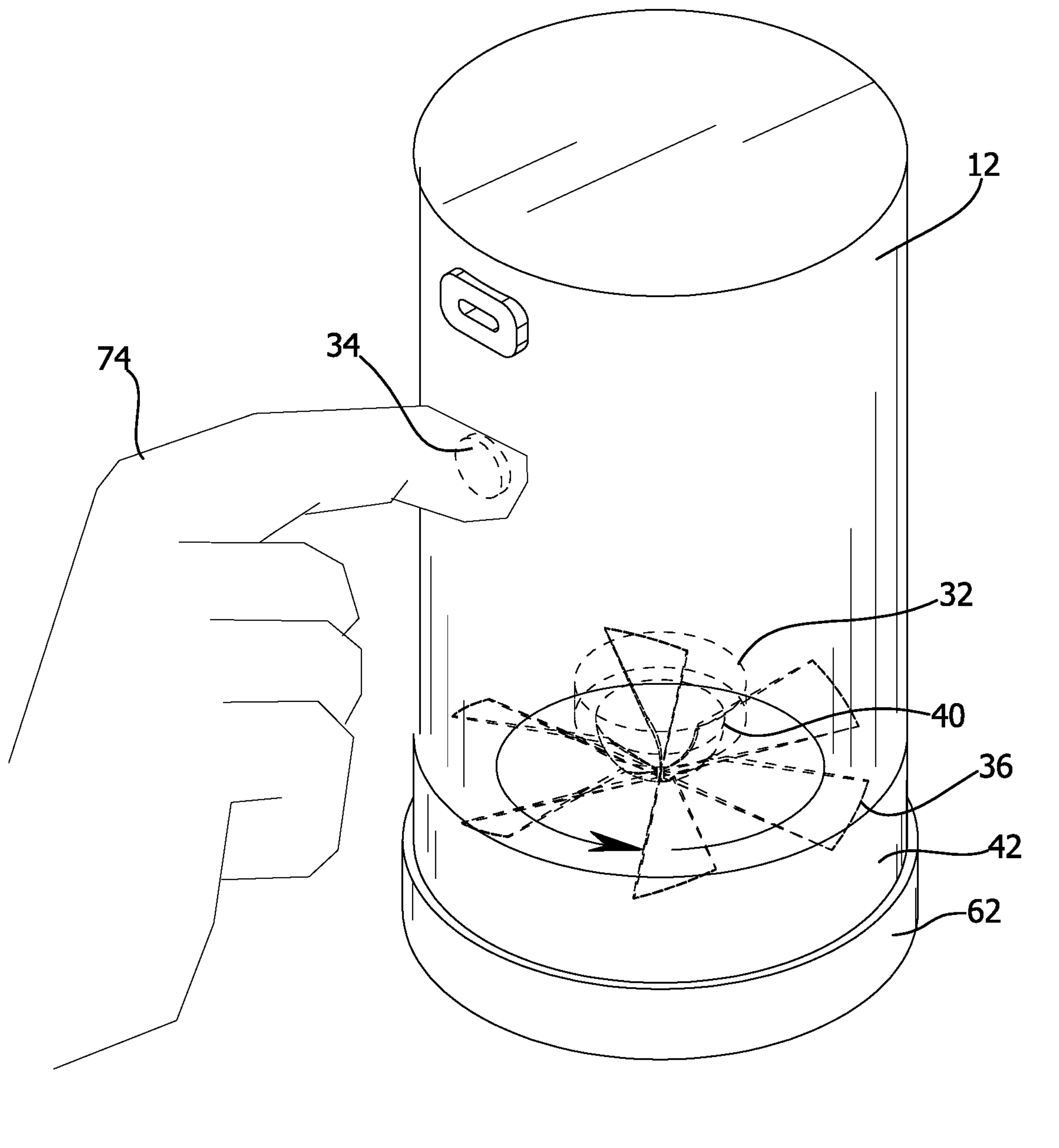


FIG. 5

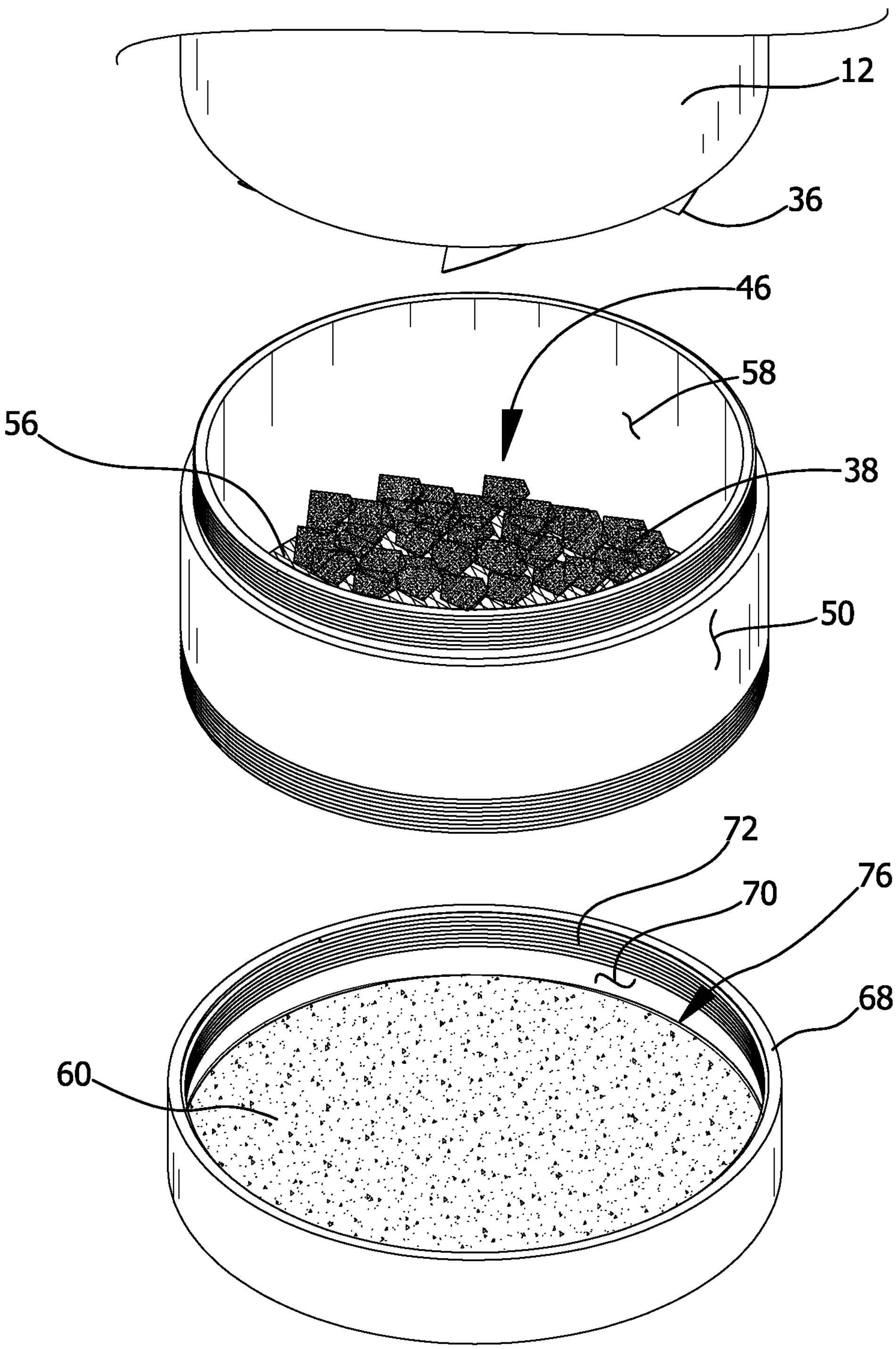


FIG. 6

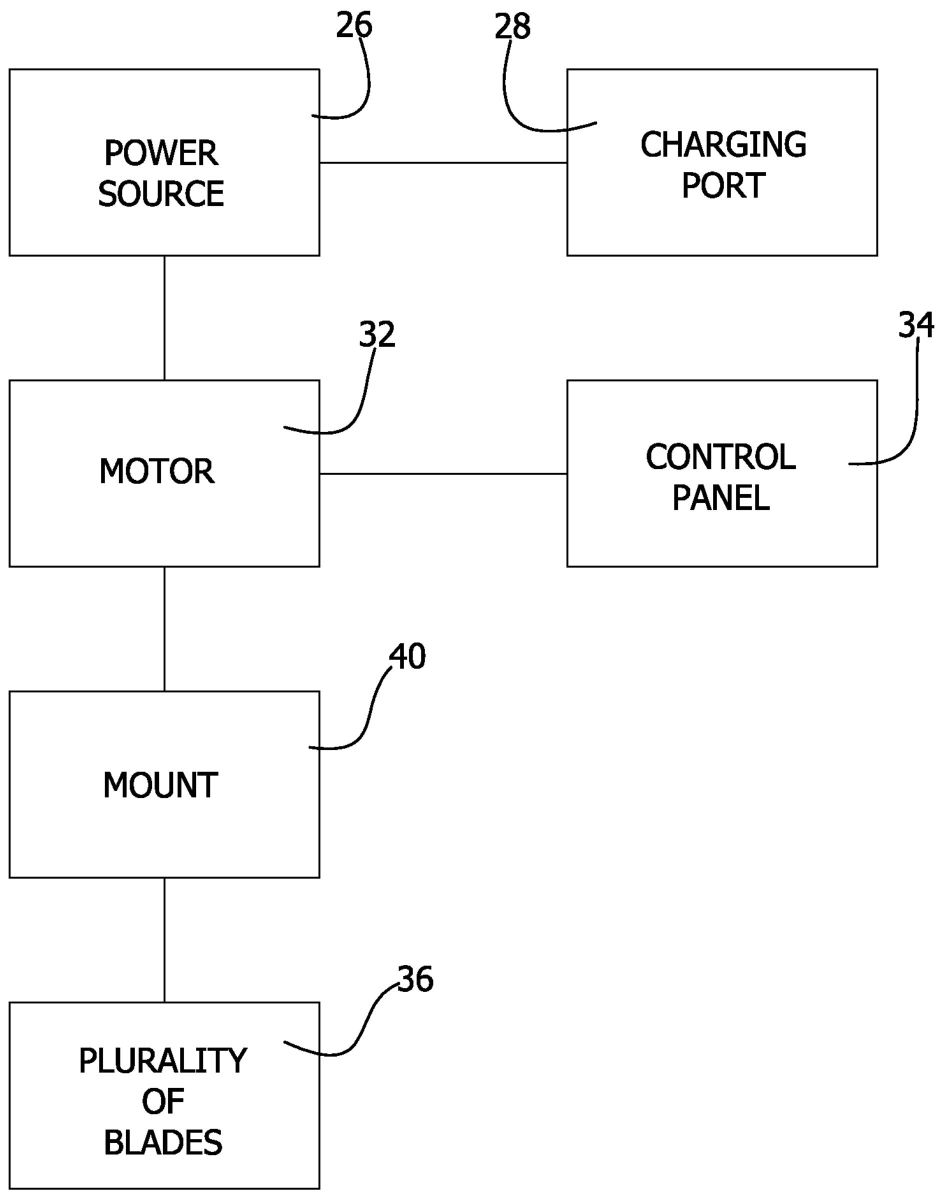

26
28
POWER SOURCE
CHARGING PORT
32
34
MOTOR
CONTROL PANEL
40
MOUNT
36
PLURALITY OF BLADES

FIG. 7

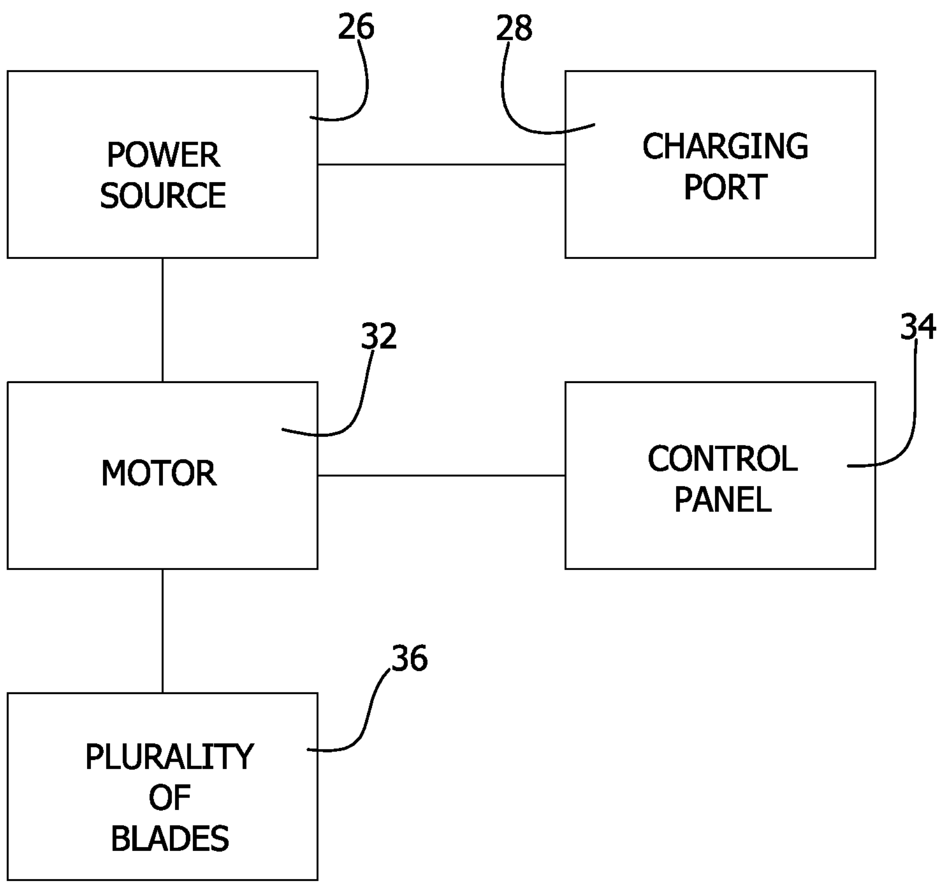

26
28
POWER SOURCE
CHARGING PORT
32
34
MOTOR
CONTROL PANEL
36
PLURALITY OF BLADES

FIG. 8

MOTORIZED GRINDER DEVICE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] Not Applicable

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0002] Not Applicable

THE NAMES OF THE PARTIES TO A JOINT RESEARCH AGREEMENT Not Applicable

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ON A COMPACT DISC OR AS A TEXT FILE VIA THE OFFICE ELECTRONIC FILING SYSTEM.

[0003] Not Applicable

STATEMENT REGARDING PRIOR DISCLOSURES BY THE INVENTOR OR JOINT INVENTOR

[0004] Not Applicable

BACKGROUND OF THE INVENTION

(1) Field of the Invention

[0005] The disclosure relates to herb grinders and more particularly pertains to a new herb grinder for grinding herbs and other plant matter into particulates with a motorized blade and sorting the particulates by size.

(2) Description of Related Art Including Information Disclosed Under 37 CFR 1.97 and 1.98

[0006] The prior art relates to herb grinders. Most herb grinders are manually operated. When a user has physical mobility concerns, such as arthritis, using a manual herb grinder can be difficult and painful. Thus, there is a need for a motorized herb grinder. The prior art does disclose some electric, motorized herb grinders. However, the prior art fails to disclose an electric herb grinder that can sort the ground particulates of the herbs by size. For example, a user might desire roughly ground herbs for one purpose, and more finely ground particulate residue for another purpose. The prior art fails to disclose a single motorized device that can remove the finely ground particulate residue from the roughly ground herbs.

BRIEF SUMMARY OF THE INVENTION

[0007] An embodiment of the disclosure meets the needs presented above by generally comprising a housing having an interior space. A power source is positioned in the interior space. A motor is positioned in the interior space. The motor is electrically coupled to the power source. A plurality of blades is coupled to the housing. The plurality of blades is coupled to the motor wherein the motor rotates the plurality of blades when the motor is actuated. The plurality of blades is sharpened wherein the plurality of blades is configured to grind an herb when the herb contacts the plurality of blades. A chamber is removably couplable to the housing. The chamber is configured to hold the herb. The plurality of blades is positioned within the chamber when the chamber is coupled to the housing wherein the plurality of blades is configured to grind the herb within the chamber when the motor rotates the plurality of blades. The chamber includes a mesh screen that is positioned within the chamber wherein the mesh screen is configured to support the herb while the herb is held in the chamber. The mesh screen is foraminous wherein the mesh screen is configured to retain the herb in the chamber when the herb is ground and wherein the mesh screen is configured to release a granulated residue of the herb from the chamber when the herb is ground. A tray is removably couplable to the chamber wherein the tray is configured to catch the granulated residue of the herb when the tray is coupled to the chamber.

[0008] There has thus been outlined, rather broadly, the more important features of the disclosure in order that the detailed description thereof that follows may be better understood, and in order that the present contribution to the art may be better appreciated. There are additional features of the disclosure that will be described hereinafter and which will form the subject matter of the claims appended hereto.

[0009] The objects of the disclosure, along with the various features of novelty which characterize the disclosure, are pointed out with particularity in the claims annexed to and forming a part of this disclosure.

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWING(S)

[0010] The disclosure will be better understood and objects other than those set forth above will become apparent when consideration is given to the following detailed description thereof. Such description makes reference to the annexed drawings wherein:

[0011] FIG. **1** is a top exploded view of a motorized grinder device according to an embodiment of the disclosure.

[0012] FIG. **2** is a bottom exploded view of an embodiment of the disclosure.

[0013] FIG. **3** is a side view of an embodiment of the disclosure.

[0014] FIG. **4** is an in-use view of an embodiment of the disclosure.

[0015] FIG. **5** is an in-use view of an embodiment of the disclosure.

[0016] FIG. **6** is an in-use view of an embodiment of the disclosure.

[0017] FIG. **7** is a block diagram view of an embodiment of the disclosure.

[0018] FIG. **8** is a block diagram view of an embodiment of the disclosure.

DETAILED DESCRIPTION OF THE INVENTION

[0019] With reference now to the drawings, and in particular to FIGS. **1** through **8** thereof, a new herb grinder embodying the principles and concepts of an embodiment of the disclosure and generally designated by the reference numeral **10** will be described.

[0020] As best illustrated in FIGS. **1** through **8**, the motorized grinder device **10** generally comprises a housing **12** with an interior space **20**. For example, the housing **12** may have a top end **14**, a bottom end **16**, and an outer wall **18** that is coupled to and extends between the top end **14** and the bottom end **16** to define an interior space **20**. The housing **12** may be cylindrical.

[0021] The housing **12** may further include a lip **22** that is coupled to and extends downwardly from the bottom end **16**. The lip **22** may have a threaded inner surface **24**.

[0022] A power source **26** is positioned in the interior space **20**. The power source **26** may include a battery. The battery may be rechargeable.

[0023] A charging port **28** may be electrically coupled to the power source **26**. The charging port **28** is designed to receive a charging cable **30** for recharging the power source **26**. The charging port **28** may be inset into the outer wall **18** of the housing **12**. The charging port **28** may be exposed within the outer wall **18** of the housing **12**. For example, the charging port **28** may be positioned adjacent to the top end **14** of the housing **12**. In certain embodiments, the charging port **28** is a micro-universal serial bus charging port **28**.

[0024] A motor **32** is positioned in the interior space **20**. The motor **32** is electrically coupled to the power source **26**.

[0025] A control panel **34** may be electrically coupled to the motor **32**. The control panel **34** actuates the motor **32** when the control panel **34** is manipulated. For example, the control panel **34** may be a button. The control panel **34** may be positioned on the outer wall **18** proximate to the top end **14** of the housing **12** to facilitate a user **74** in manipulating the control panel **34** while the user **74** is holding the housing **12**.

[0026] A plurality of blades **36** is coupled to the housing **12**. The plurality of blades **36** is generally rotatably coupled to the bottom end **16** of the housing **12**. The plurality of blades **36** is coupled to the motor **32** wherein the motor **32** rotates the plurality of blades **36** when the motor **32** is actuated. The plurality of blades **36** is sharpened wherein the plurality of blades **36** is designed to grind an herb **38** when the herb **38** contacts the plurality of blades **36**.

[0027] Some embodiments include a mount **40** that rotatably couples the plurality of blades **36** to the bottom end **16** of the housing **12**. The mount **40** may be rotatably coupled to the housing **12** wherein the motor **32** rotates the mount **40** to rotate the plurality of blades **36**. An example of such embodiments is provided in FIG. **7**. Alternatively, the mount **40** may be immovably coupled to the housing and the motor may rotate the plurality of blades directly. An example of such an embodiment is provided in FIG. **8**.

[0028] The mount **40** generally extends outwardly from the bottom end **16** of the housing **12** wherein the mount **40** is designed to space the plurality of blades **36** from the housing **12**. For example, the mount **40** may have a hemispherical shape. The mount **40** may be centrally positioned on the bottom end **16** of the housing **12** wherein the mount **40** is designed to center the plurality of blades **36** relative to the bottom end **16** of the housing **12**.

[0029] A chamber **42** is removably couplable to the housing **12**. The chamber **42** is designed to hold the herb **38**. The mount **40** and the plurality of blades **36** may be positioned within the chamber **42** when the chamber **42** is coupled to the housing **12** wherein the plurality of blades **36** is designed to grind the herb **38** within the chamber **42** when the motor **32** rotates one or both of the mount **40** and the plurality of blades **36**.

[0030] The chamber **42** may further include an upper edge **44** that is positioned adjacent to the housing **12** when the chamber **42** is coupled to the housing **12**. The upper edge **44** generally defines an opening **46** into the chamber **42**. The opening **46** is generally configured to facilitate the user **74** in placing the herb **38** within the chamber **42**. The upper edge **44** may be coupled to the lip **22** of the housing **12** when the chamber **42** is coupled to the housing **12**.

[0031] A lower edge **48** of the chamber is spaced from the upper edge **44**. An exterior surface **50** extends between the upper edge **44** and the lower edge **48**. The exterior surface **50** may further include an upper chamber threading **52** extending downwardly from the upper edge **44** toward the lower edge **48**. A lower chamber threading **54** may extend upwardly along the exterior surface **50** from the lower edge **48** toward the upper edge **44**. The lower chamber threading may be spaced from the upper chamber threading **52**, as shown in FIGS. **1** and **6**.

[0032] A mesh screen **56** is coupled to an interior surface **58** of the chamber **42**. The mesh screen **56** is generally positioned within the chamber **42** adjacent to the lower edge **48** wherein the mesh screen **56** is designed to support the herb **38** while the herb **38** is held in the chamber **42**. The mesh screen **56** is foraminous wherein the mesh screen **56** is designed to retain the herb **38** in the chamber **42** when the herb **38** is ground and wherein the mesh screen **56** is designed to release a granulated residue **60** of the herb **38** from the chamber **42** when the herb **38** is ground.

[0033] The upper chamber threading **52** is generally complementary to the threaded inner surface **24** of the lip **22** of the housing **12** wherein the chamber **42** is threadably couplable to the housing **12**.

[0034] A tray **62** is removably couplable to the chamber **42**. The tray **62** may have a base wall **64** and a peripheral wall **66** that is coupled to and extends upwardly from the base wall **64** wherein the tray **62** is designed to catch the granulated residue **60** of the herb **38** when the tray **62** is coupled to the chamber **42**.

[0035] The peripheral wall **66** may have a free edge **68** that is spaced from the base wall **64**. The free edge **68** may define an orifice **76** that is designed to facilitate access to the granulated residue **60** when the tray **62** is removed from the chamber **42**. An interior side **70** of the peripheral wall **66** may have a tray threading **72** extending downwardly from the free edge **68** toward the base wall **64**. The tray threading **72** is complementary to the lower chamber threading **54** on the exterior surface **50** of the chamber **42** wherein the tray **62** is threadably couplable to the chamber **42**.

[0036] In use, the user **74** can place the herb **38** onto the mesh screen **56** within the chamber **42**. The user **74** can threadably couple the housing **12** and the tray **62** to opposing lateral ends of the chamber **42**. More specifically, the lip **22** of the housing **12** can be coupled to the upper chamber threading **52** extending from the upper edge **44** of the chamber **42**, and the peripheral wall **66** of the tray **62** can be coupled to the lower chamber threading **54** extending from the lower edge **48** of the chamber **42**. The tray **62** is thus positioned directly beneath the mesh screen **56** to catch the granulated residue **60** of the herb **38** as the herb **38** is being ground by the plurality of blades **36**. The user **74** can use the control panel **34** to actuate the motor **32** to rotate the plurality of blades **36** and grind the herb **38**. The ground herb **38** can then be removed from the motorized grinder device **10** by detaching the chamber **42** from the housing **12**. To access the granulated residue **60**, the user **74** can detach the tray **62** from the chamber **42**.

[0037] With respect to the above description then, it is to be realized that the optimum dimensional relationships for the parts of an embodiment enabled by the disclosure, to include variations in size, materials, shape, form, function and manner of operation, assembly and use, are deemed readily apparent and obvious to one skilled in the art, and all equivalent relationships to those illustrated in the drawings and described in the specification are intended to be encompassed by an embodiment of the disclosure.

[0038] Therefore, the foregoing is considered as illustrative only of the principles of the disclosure. Further, since numerous modifications and changes will readily occur to those skilled in the art, it is not desired to limit the disclosure to the exact construction and operation shown and described, and accordingly, all suitable modifications and equivalents may be resorted to, falling within the scope of the disclosure. In this patent document, the word "comprising" is used in its non-limiting sense to mean that items following the word are included, but items not specifically mentioned are not excluded. A reference to an element by the indefinite article "a" does not exclude the possibility that more than one of the element is present, unless the context clearly requires that there be only one of the elements.

I claim:

1. An electric herb grinding assembly comprising:

a housing having an interior space;

a power source being positioned in the interior space;

a motor being electrically coupled to the power source;

a plurality of blades being coupled to the housing, the plurality of blades being coupled to the motor wherein the motor rotates the plurality of blades when the motor is actuated, the plurality of blades being sharpened wherein the plurality of blades is configured to grind an herb when the herb contacts the plurality of blades;

a chamber being removably couplable to the housing, the chamber being configured to hold the herb, the plurality of blades being positioned within the chamber when the chamber is coupled to the housing wherein the plurality of blades is configured to grind the herb within the chamber when the motor rotates the plurality of blades, the chamber further comprising:

a mesh screen being positioned within the chamber wherein the mesh screen is configured to support the herb while the herb is held in the chamber, the mesh screen being foraminous wherein the mesh screen is configured to retain the herb in the chamber when the herb is ground and wherein the mesh screen is configured to release a granulated residue of the herb from the chamber when the herb is ground; and

a tray being removably couplable to the chamber wherein the tray is configured to catch the granulated residue of the herb when the tray is coupled to the chamber.

2. The electric herb grinding assembly of claim **1**, further comprising:

a lip being coupled to and extending downwardly from the bottom end of the housing, the lip having a threaded inner surface; and

an upper chamber threading extending downwardly from an exterior surface of an upper edge of the chamber, the upper chamber threading being complementary to the threaded inner surface of the lip of the housing wherein the chamber is threadably couplable to the housing.

3. The electric herb grinding assembly of claim **1**, further comprising:

a lower edge of the chamber being positioned adjacent to the mesh screen, the lower edge having a lower chamber threading extending upwardly from an exterior surface of the lower edge;

a tray threading extending downwardly from a free edge of the tray, the tray threading being complementary to the lower chamber threading on the exterior surface of the chamber wherein the tray is threadably couplable to the chamber.

4. The electric herb grinding assembly of claim **1**, further comprising a charging port being electrically coupled to the power source, the charging port being configured to receive a charging cable for recharging the power source.

5. The electric herb grinding assembly of claim **1**, the charging port being inset into an outer wall of the housing, the charging port being exposed within the outer wall of the housing.

6. The electric herb grinding assembly of claim **1**, further comprising a mount rotatably coupling the plurality of blades to a bottom end of the housing.

7. The electric herb grinding assembly of claim **6**, wherein the mount is rotatably coupled to the housing wherein the motor rotates the mount to rotate the plurality of blades.

8. The electric herb grinding assembly of claim **6**, wherein the mount is centrally positioned on the bottom end of the housing wherein the mount is configured to center the plurality of blades relative to the bottom end of the housing.

9. The electric herb grinding assembly of claim **6**, wherein the mount extends outwardly from the bottom end of the housing wherein the mount is configured to space the plurality of blades from the housing.

10. The electric herb grinding assembly of claim **1**, further comprising a control panel being electrically coupled to the motor, the control panel actuating the motor when the control panel is manipulated.

11. The electric herb grinding assembly of claim **10**, wherein the control panel is positioned on the housing.

12. An electric herb grinding assembly comprising:

a housing having a top end, a bottom end, and an outer wall being coupled to and extending between the top end and the bottom end to define an interior space, the housing being cylindrical, the housing further comprising:

a lip being coupled to and extending downwardly from the bottom end, the lip having a threaded inner surface;

a power source being positioned in the interior space, the power source comprising a battery, the battery being rechargeable;

a charging port being electrically coupled to the power source, the charging port being configured to receive a charging cable for recharging the power source, the charging port being inset into the outer wall of the housing, the charging port being exposed within the outer wall of the housing, the charging port being positioned adjacent to the top end of the housing, the charging port being a micro-universal serial bus charging port;

a motor being positioned in the interior space, the motor being electrically coupled to the power source;

a control panel being electrically coupled to the motor, the control panel actuating the motor when the control panel is manipulated, the control panel being positioned on the outer wall proximate to the top end of the housing;

a plurality of blades being coupled to the housing, the plurality of blades being rotatably coupled to the bottom end of the housing, the plurality of blades being coupled to the motor wherein the motor rotates the plurality of blades when the motor is actuated, the plurality of blades being sharpened wherein the plurality of blades is configured to grind an herb when the herb contacts the plurality of blades;

a mount rotatably coupling the plurality of blades to the bottom end of the housing, the mount being rotatably coupled to the housing wherein the motor rotates the mount to rotate the plurality of blades, the mount being centrally positioned on the bottom end of the housing wherein the mount is configured to center the plurality of blades relative to the bottom end of the housing, the mount extending outwardly from the bottom end of the housing wherein the mount is configured to space the plurality of blades from the housing, the mount having a hemispherical shape;

a chamber being removably couplable to the housing, the chamber being configured to hold the herb, the mount and the plurality of blades being positioned within the chamber when the chamber is coupled to the housing wherein the plurality of blades is configured to grind the herb within the chamber when the motor rotates the mount, the chamber further comprising:

an upper edge being positioned adjacent to the housing when the chamber is coupled to the housing, the upper edge defining an opening into the chamber, the upper edge being coupled to the lip of the housing when the chamber is coupled to the housing;

a lower edge being spaced from the upper edge;

an exterior surface extending between the upper edge and the lower edge, the exterior surface further comprising:

an upper chamber threading extending downwardly from the upper edge toward the lower edge;

a lower chamber threading extending upwardly from the lower edge toward the upper edge, the lower chamber threading being spaced from the upper chamber threading;

a mesh screen being coupled to an interior surface of the chamber, the mesh screen being positioned within the chamber adjacent to the lower edge wherein the mesh screen is configured to support the herb while the herb is held in the chamber, the mesh screen being foraminous wherein the mesh screen is configured to retain the herb in the chamber when the herb is ground and wherein the mesh screen is configured to release a granulated residue of the herb from the chamber when the herb is ground;

the upper chamber threading being complementary to the threaded inner surface of the lip of the housing wherein the chamber is threadably couplable to the housing; and

a tray being removably couplable to the chamber, the tray having a base wall and a peripheral wall being coupled to and extending upwardly from the base wall wherein the tray is configured to catch the granulated residue of the herb when the tray is coupled to the chamber, the peripheral wall having a free edge being spaced from the base wall, the free edge defining an orifice wherein the free edge is configured to facilitate access to the granulated residue when the tray is removed from the chamber, an interior side of the peripheral wall having a tray threading extending downwardly from the free edge toward the base wall, the tray threading being complementary to the lower chamber threading on the exterior surface of the chamber wherein the tray is threadably couplable to the chamber.

* * * * *